

OpenRiverCam in Indonesia: What the Pilot Taught Us, and the Path Forward

Indonesia ORC Deployment — Spring 2026

PMI · IPB · BHLK · American Red Cross

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Contents

The pilot, and what we think comes next	3
Why a low-cost station is worth studying	4
What is being offered for study	4
What four months in the field taught us	6
Eleven things we would do differently	8
Building and looking after the units in Indonesia	12
Hosting the data in Indonesia	12
Choosing the sites	12
Working together	13
Meeting the conditions for the data to be accepted	13
Questions for consideration	14
What we can offer from here	15
Supporting documentation	15

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Partners: Palang Merah Indonesia · Institut Pertanian Bogor · Balai Hidrologi dan Lingkungan Keairan · American Red Cross

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Reading time: about twelve minutes.

Companion document: Appendix — Recommendations for Replication of the OpenRiverCam Station Design, written for the engineers who will build, procure and operate the units. Every measurement, interface specification and procedure referred to below is held there.

The pilot, and what we think comes next

The pilot has been a good experience. It put a station in the water, gave us four months of behaviour to learn from, and produced something more valuable than either: a working relationship between PMI, IPB and BHLK that did not exist before. Colleagues who had not worked together now share a problem, a site and a vocabulary for both. We would do it again.

Our recommendation is that the next step is yours. Not because the partnership has run its course, but because of what the pilot actually produced. It produced knowledge rather than a body of measurements, and knowledge is worth more to you as an input to your own approach than as a set of parts to copy. We would rather help you start your own design than hand you ours to maintain.

This document is that knowledge, offered in full and including the parts that reflect badly on choices we made. It is written to give an accurate basis for your decision rather than an encouraging one.

At Sukabumi on 21 August, BHLK offered to build one to three stations as a pilot, offered server capacity for the data, and recommended moving the present site to open ground. We welcome all three offers, and this document responds to them and to the request to study the current design.

The scope is the technology only. We describe the station as built, what four months in the field revealed about it, and what we would do differently. We do not assess what the resulting data is fit to support, or what accuracy any particular application requires. Those judgements belong with IPB, BHLK and their federal partners. Where a measurement requirement appears below, it is one recorded from you rather than proposed by us.

The evidence is thin, and we ask you to treat it as thin. One station, on one river, observed across part of one season. A second station was built but never installed. The

calibration the station runs on today was recovered from a survey that failed. What the deployment has produced so far is knowledge about the design, not a body of measurements.

Why a low-cost station is worth studying

A river gauge is only useful if there are enough of them. How many stations an agency can afford determines how much of a catchment it can observe, and that is a different question from how good any single station is.

The station described here came to **USD 1,340 in materials**. That figure is the electronics and enclosure only: Sukabumi already had a 200 W panel and a 50 Ah battery on site, so no solar array is included in it, and a new solar site would have to add one. A mains site does not. The lowest-cost automatic water-level station we could confirm on the Indonesian government e-catalogue is the domestically assembled IDDATA RL03 with GSM telemetry, at **Rp 58,000,000, approximately USD 3,600 before VAT**, and it measures stage only. The comparison is not like for like: the RL03 is a supported commercial instrument, and this is a pilot-stage assembly. The reason it is worth stating is that at this price a basin authority can consider a network where it might otherwise consider a single gauge.

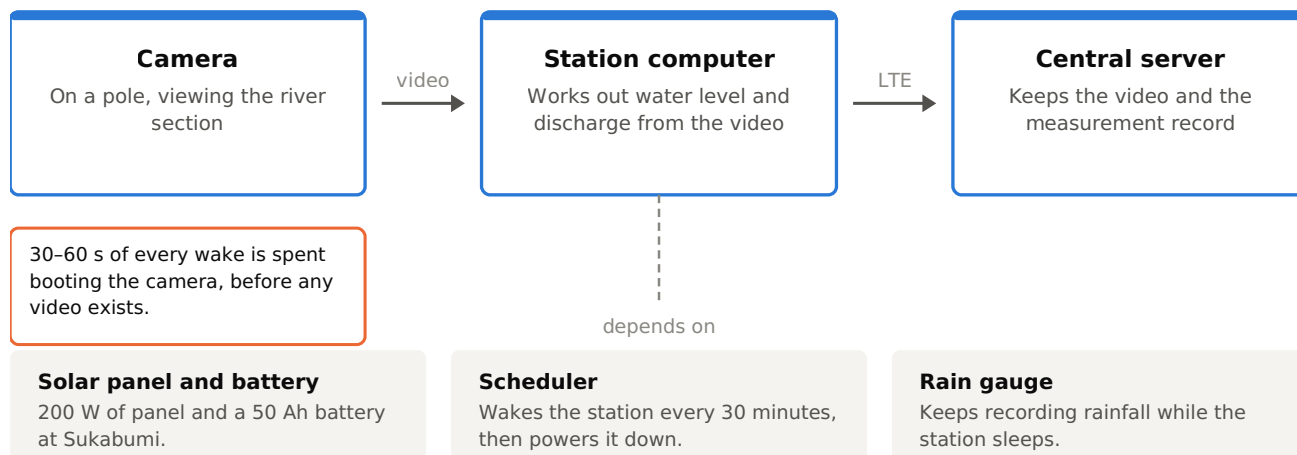
That possibility is the whole of the argument for studying the design. Everything else in this document is about whether the thing can be made reliable enough to deserve it.

What is being offered for study

A camera on a pole watches the river. A small computer in a weatherproof box turns the video into a water level and a discharge figure, and a mobile data link sends both, with the video, to a server. At Sukabumi the station runs on solar power and wakes on a thirty-minute cycle: it starts, captures, processes, uploads, and shuts down again.

Figure 1 How a measurement is made

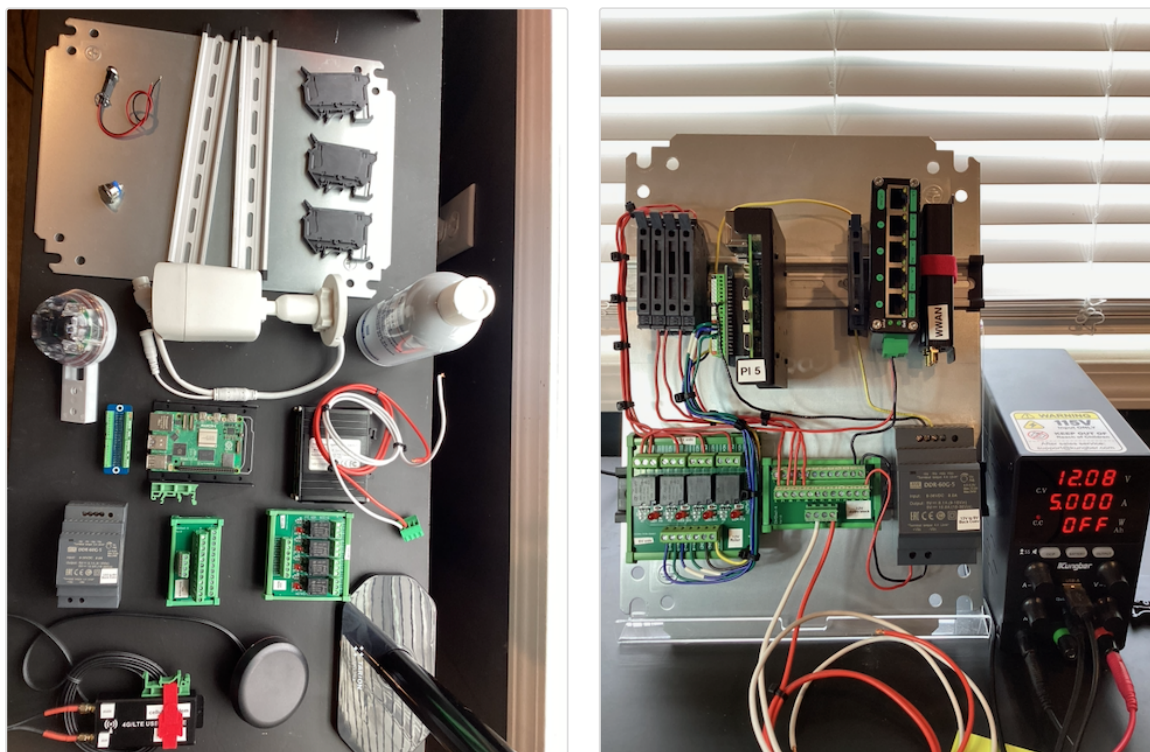
The path a single five-second video takes, from the river to the record.



The station is awake about two minutes in every thirty. Everything above happens inside that window.

The camera on its pole sends video to the station computer, which derives water level and discharge; both travel over the mobile network to the server. The three things the computer depends on are shown beneath it. The whole sequence happens inside a waking period of about two minutes, of which the first thirty to sixty seconds is spent waiting for the camera to start.

Five constraints governed every choice of part, and they are the reason the design can be copied at all: commodity electronics with more than one supplier; no soldering, with every connection made by screw terminal, plug or header; no specialist assembly skills; common hand tools; and any part replaceable in five minutes.



The parts for one station, and the same parts wired onto the mounting plate under bench power. Nothing here is fabricated, and nothing is single-source. A technician with a screwdriver can replace any of it.

Those constraints are what make local repair possible, and local repair is what keeps a station working in a place where the nearest specialist is a day away. We would ask you to hold them even where they are inconvenient.

What four months in the field taught us

Three design gaps matter more than the rest. Each is something the station should have been able to do and could not, and for each we say what we would build instead. The measurements behind them are in the appendix, recorded there as known failure modes of this design rather than as a performance record.

The station cannot report its own condition. When it stopped, it stopped quietly. Nothing in the design let an interruption announce itself, so periods when it was not running had to be reconstructed afterwards by querying the server database. A fault the system never reports cannot be found by watching for it, and no amount of attention from the field makes up for a machine that cannot say what is wrong. There is a related gap: a maintenance setting that suppresses recording and holds the processor awake has no time limit and raises no alarm, so it can stay on without anyone being told.

What we would build instead: the station reports its own state on every waking, and any mode that suppresses data or raises energy use expires on its own and raises an alert while

it is set. Health reporting belongs in the requirements alongside the measurement, not added afterwards when something has already gone quiet. That is R4 and R5.

Nothing reconciles what was recorded against what arrived. The station knows what it captured and the server knows what it received, and no part of the design compares the two. Video and sensor readings also travel by separate paths and fail separately, so neither confirms the other. Data can go missing without producing a symptom at either end, which is the kind of loss that is found late or not at all.

What we would build instead: a scheduled comparison of the station's own record against the server's, raising a difference as an alert. It is a small piece of software and it should have been in the first version. That is R7.

Water level is read from the image, and daylight defeats it. Sukabumi has no water-level sensor and derives the level from the video. In a sample of 200 captures every rejection fell between 06:00 and 19:00, and none at night. Because the water level is worked out first, a daytime rejection costs the whole discharge measurement and not only the level.

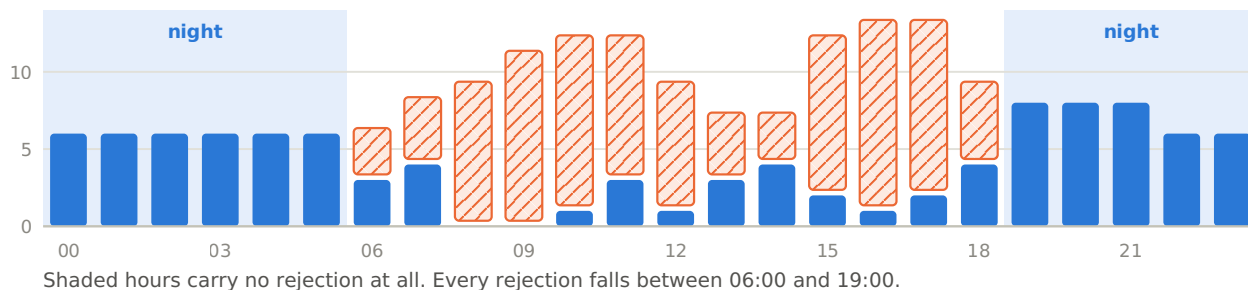
What we would build instead: an independent water-level reference — a sensor, or a staff gauge inside the camera's view — so the measurement does not depend on a single optical method succeeding. Deriving the level from the image is a good capability to have and a poor one to rely on alone. That is R1, and it is the change with the clearest benefit.

Figure 2 Reading the water level from video, 8 to 14 July 2026

200 captures. A rejection costs the whole discharge measurement, not only the water level.

■ Water level accepted (100) ■ Rejected — not confident enough (100)

A By hour of day (local time, WIB)



B How confident the reading was, for all 200 captures



Rejected captures are not near misses: half sit at 1.63 or below, and only 23 of the 100 reach 1.8.
Accepted captures spread well above the threshold, though 36 of them sit between 2.0 and 3.0.

Captures counted by hour of day, and the confidence score of all 200 captures against the threshold at which a water level is accepted. Rejections peak in mid-morning and again in mid-afternoon, with fewer in the early afternoon — the pattern a sun-angle effect produces and a general brightness effect would not.

The cause is most likely reflected sunlight, and that reading is well supported by the shape of the day, but it is not settled. The recommendation below rests on the failure, which is measured, and not on the explanation.

Eleven things we would do differently

These are offered as input to your design rather than as corrections to ours. Some will not apply to the approach you choose, and we would not expect them to be adopted as a set. The appendix gives the observation behind each and the detail needed to act on it. They fall into three groups, and the first two decide whether a station produces information anyone can use.

Let the station account for itself. A station that cannot report its own condition cannot be looked after from a distance, and travelling to a river to learn whether a computer is running is an expensive way to find out. Nothing in this group is difficult or costly. The

maintenance setting described above was readable remotely the whole time, and no software ever asked. **R4, R5, R6, R7.**

Make the measurement trustworthy. Two changes carry most of the weight here. The station needs an independent water-level reference (**R1**) — a sensor, or a staff gauge inside the camera’s view tied to the same papan duga air zero BBWS uses — because without one it measures dependably only at night.

And the survey should be planned as professional work from the start (**R2**), because the surveyed shape of the riverbed is an input the processing cannot recover. We learned this the hard way, and IPB fixed it. Two RTK surveys at Sukabumi on consecutive days, with the same equipment and the same crew, disagreed with themselves by about **99 cm horizontally and 139 cm vertically**, and nothing in our workflow caught it on site. **IPB’s total-station survey replaced that approach and is what the station runs on today** — a camera configuration built from IPB data alone, fitted at **3.7 cm RMSE** against a 5 cm target. The lesson for a new site is to plan for that standard of survey from the beginning rather than arriving at it after two failures, and to carry an independent field check of a different kind so a survey problem is caught before people go home.

Make the units buildable and maintainable in Indonesia. BHLK will buy in Indonesia under Indonesian procurement rules, so substitution is the expected case rather than a risk to be managed. Documenting which properties bind and which do not (**R3**) is what allows that without every substitution becoming a question referred back to us. **R8 and R11** change where the equipment lives and how it is powered, and both make the station easier to keep running; **R9 and R10** are about how parts are chosen and one board fewer.

	Change	What it buys	Cost
R1	Fit an independent water-level reference	Measurement through the day, not only at night	A level sensor, or a staff gauge in view
R2	Commission a professional survey first	The one input the processing cannot recover	Rp 5–15 million per site
R3	Specify interfaces, not part numbers	Local procurement without referring back to us	Documentation only
R4	Health reporting and mode alarms as requirements	The station reports its own condition	Negligible
R5	Station sends diagnostics; no login required	Support that works inside a short waking period	Small
R6	Record voltage and current together	Separates a failing battery from a heavy load	A current-sensing module
R7	Compare recorded against received, automatically	Loss is noticed rather than discovered later	Negligible

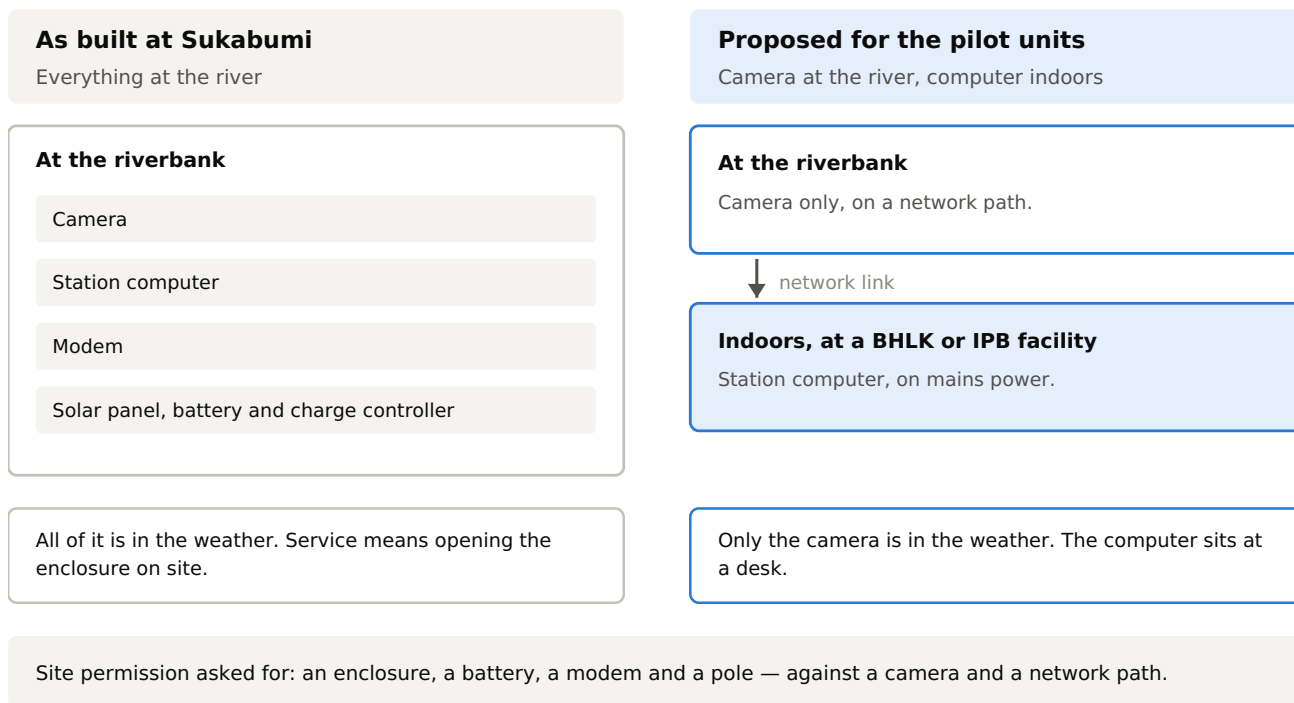
	Change	What it buys	Cost
R8	Put the computer indoors	Removes it from heat, humidity, dust and travel	Designed, not yet field tested
R9	Budget per station; check the interface before buying	Keeps a network affordable without hidden costs	Screening effort
R10	Use the computer's own clock where the site allows	One less board to fail	Saves about USD 50 per station
R11	Build on mains and leave it running, where continuous reporting is wanted	Removes most of the gaps above	No solar array to buy; constrains siting

R8 deserves a note, because it is the least proven and possibly the most useful.

Camera and sensors stay at the river; the computer runs at a BHLK or IPB facility over a network link. That reduces the work at the riverbank to mounting a camera, puts the computer where temperature, humidity, dust and access are controlled, and lowers what has to be asked of a landowner. Where the field unit is a standard security camera, installation and support also fall within a supply chain that already exists across Indonesia. Stated plainly: this arrangement is designed and **not yet field tested**. We offer it as the pilot's first experiment, not as a proven alternative.

Figure 3 Two arrangements of the same design

Recommendation R8. The right-hand arrangement is designed but not yet field tested.

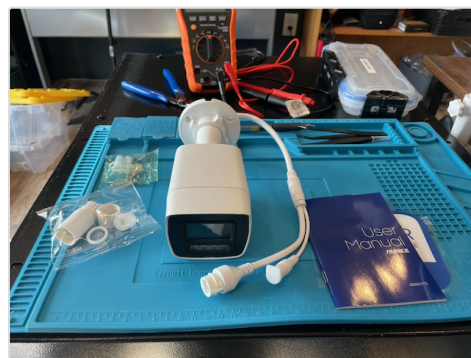


As built, the camera, computer, modem and power system all sit in one enclosure at the riverbank, and all of it is in the weather. As proposed for the pilot units, only the camera stays at the river. The right-hand arrangement is designed and not yet field tested.

One lesson about method

The budget was applied part by part: for each function, the cheapest item meeting the requirement. That produced a working station at USD 1,340, and it has one failure this deployment demonstrated. **It prices each part against its specification sheet, not against what that part's limits cost the rest of the system.**

The camera is the clear case. At about USD 60 against a professional alternative near USD 1,268, it was not really a choice. The unit meets every line of its specification and still costs the system a fifth of its video quality, a slower measurement cycle, and thirty to sixty seconds of battery on every waking — none of which appeared in the comparison that selected it. Its light also fires at full brightness whenever it starts and cannot be turned off, which at the present site is forty-eight flashes a day. That is a reason to weigh siting and cycle length, and not only the power budget.



Capable hardware, running a reseller's version of the manufacturer's software. Every limitation we met came from that software rather than from the optics or the sensor.

Building and looking after the units in Indonesia

The design was made to be built by people who are not electronics specialists, with tools they already own. That intent only becomes real if the parts can be bought locally and the documentation supports substitution, which is what R3 addresses.

For maintenance, the pattern that has worked is a small stock of spares held at the local PMI chapter, so a failure is a part swap rather than a shipment. Where the field unit is a standard security camera (R8), installation and support fall within a supply chain that already exists across Indonesia, with a deeper set of suppliers than specialist hydrometric equipment.

We would rather help build the capacity to maintain these stations than remain the place they are sent when they break.

Hosting the data in Indonesia

BHLK's offer of server capacity suits the pilot well, and holding the data in Indonesia carries a clear benefit for a government-partnered deployment. The server software is packaged for straightforward installation. Two constraints are worth designing for rather than discovering: the video store and the database must share one filesystem, so plan storage as a single volume; and the server and station software versions are tied together, so a server upgrade obliges the stations to follow, which a remote station cannot do on demand.

We would suggest running the BHLK instance alongside the existing one, receiving the same data, until it has completed a full operating cycle including an upgrade. It would also be worth agreeing in writing, in advance, where the authoritative copy sits, who administers it, who has access, and what the retention policy is.

Choosing the sites

BHLK's recommendation to move to open ground free of obstruction is supported, and the field record adds independent evidence for it. An open site helps with three separate problems at once: satellite positioning is degraded by nearby buildings, which is among the leading candidate causes of the survey trouble at the present site; the angle between sun, water and camera can be chosen to avoid the daytime failure through more of the day; and the view across the section determines how much of the flow the camera can resolve.

Two things to plan for. A move costs a fresh survey and a fresh calibration, not only a physical relocation, and the survey is the expensive part. And site permission should be settled before a unit is built for a particular site: a complete station was built and tested for an intended Jakarta site whose permission was expected and fell through during the April visit. We would not build to a specific site again before being told the permission is in place.

That station is still available, and we would suggest a particular use for it. It is complete and software-ready, held at Wisma PMI in Jakarta, and has not been powered on since April. The plan had been to transfer it to IPB to install at a river.

We would suggest it serves you best as a study and test unit: a complete working example to open, trace, power up, take apart and rebuild while your own units are built — and, if that is useful, installed somewhere convenient and local so it can be exercised against real water. That answers the request recorded at the meeting, access to the current design in order to study it, with hardware rather than only with documents.

What we would not recommend is putting it into service as an operational station with expectations of availability and consistent data. It was built to the design this document recommends changing, so it carries the problems described above. Those are acceptable in something you are learning from, and a poor foundation for a record anyone depends on.

Working together

The division recorded at the meeting matches what this project's experience supports. Technology development and field operations are different disciplines, and where the responsibilities blur, field problems land in research inboxes and wait.

- **BHLK** — data processing, conformance with standards, and the route to acceptance within PUPR data systems. Offers server capacity.
- **IPB** — design, calibration methodology, training material, and the development path for future sensor types.
- **PMI** — user of the information, and the operational side: installation and maintenance, siting within its mission, response when something fails, and spares held at the local chapter.

Two suggestions offered for your consideration. Writing the split into the collaboration agreement, rather than holding it as an understanding between the people currently working together, so that it survives staff changes on any side. And keeping a light joint forum — a regular call, or a shared issue list — for the cases that do not sort cleanly into one of the three roles.

The commitments this implies for PMI have not been discussed with PMI National Headquarters. They are recorded here as the meeting recorded them, not as an agreed position.

Meeting the conditions for the data to be accepted

We reproduce the conditions the BBWS record is kept to because they constrain the design, the parts list and the installation: a unit lacking the necessary hardware cannot meet them

afterwards. Whether the output should be accepted, and on what terms, is not ours to assess.

Stage in metres above local datum, referenced to the same papan duga air zero, at 1 cm resolution or better. A time step of 15 minutes at minimum and 5 minutes preferred for SIH3 and SIHLSDA ingest; the station as built reports every 30 minutes. Discharge in m³/s with uncertainty documented following **SNI 8066:2015** principles, or WMO-No. 168 Chapter 5. A standard transfer format carrying value, timestamp and station identifier. Paired daily manual readings against the co-located staff gauge during commissioning. A site report recording coordinates, local benchmark, gauge zero elevation, sensor height and the calibration record.

Documenting uncertainty is the condition that bears hardest on the equipment. A steady offset can be corrected afterwards. Error that varies from point to point across the section cannot: there is no single correction factor, and it enters the uncertainty rather than being removed from it. That is why R2 and R1 are not separable from these conditions, and why neither can be added cheaply once a unit is installed.

Questions for consideration

We offer these as open, because they are.

- Who should hold the Jakarta station as a study and test unit? It was to transfer to IPB for deployment. Whether IPB or BHLK is better placed to hold it is a question for the two of you, and we will support either.
- Can a staff gauge be read from the camera image accurately enough? If it can, R1 needs no separate sensor and the station aligns directly with BBWS practice.
- Two interruptions remain unexplained. They carry the energy signature of the maintenance setting without that setting being on, which suggests something else can make capture fail in the same way.
- Whether the recovery-voltage threshold bounds how long an interruption lasts. One recovered unattended in 6.5 hours against a prior range of 21 hours to 9.3 days. One observation is not a result, and a competing reading is that the threshold may hold the station off rather than bring it back.
- How the velocity measurement performs across the surface conditions the pilot sites present. Not characterised at Sukabumi; it should be measured rather than assumed.
- Absolute discharge accuracy at Sukabumi, unresolved pending the survey and not resolvable without it.

We would welcome your reading of these, and we expect your work to change some of what is written above. That is the point of handing it over rather than handing it down.

What we can offer from here

We are not proposing to build your stations. What we can offer is the record: this document, the appendix behind it, the operator and assembly documentation in English and Bahasa Indonesia, the software, and the built station at Wisma PMI to take apart. Beyond that, whatever is useful — reviewing a design, answering a question about something that surprised us, looking at data that does not behave.

Sukabumi will keep running, and we will keep reporting what it does, including the parts that go wrong. If that is useful to your design work, it is yours.

The pilot did what a pilot should do. It found the problems while they were still cheap, and it introduced three organisations to one another. The next station in this story should be one you designed.

Supporting documentation

Measurements, interface specifications and procedures are in the companion appendix: the camera firmware limitations and capture path (A1); firmware replacement risks (A2); survey scope of work, acceptance checks and contract terms (A3); the availability record and maintenance-mode statistics (A4); the water-level dataset (A5); data delivery measurements (A6); and the power, scheduling and always-on comparisons (A7).

Operator and assembly documentation, in English and Bahasa Indonesia, is available on request.